
PROJECT REPORT

Student Mental Health & Counselling Support System

DBMS Project Report



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

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Submitted by:

1. Aadish Jain – 1024030496
2. Raghu Kohli – 1024030500

Under the Guidance of: Respected Faculty

Dr. Reaya Grewal

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3. INTRODUCTION

3.1 Background of the Problem

Mental health is an increasingly critical concern among college and university students globally. The academic rigor, peer pressure, career anxieties, and personal challenges often culminate in severe psychological distress, leading to anxiety, depression, and burnout. Educational institutions strive to provide on-campus counselling centers to address these issues.

However, the operational side of managing student mental health records is fraught with inefficiencies. Most university counselling centers still rely on disjointed electronic spreadsheets, basic calendar applications, or paper-based records to manage highly sensitive appointments, therapy session notes, and diagnostic assessments. This fragmented approach leads to scheduling conflicts, lost data, and crucially, the inability to identify high-risk students in a timely manner.

3.2 Motivation for Choosing the Project

The primary motivation behind this project is to bridge the gap between mental health care and modern database technology. Having observed the administrative bottlenecks that university counsellors face—spending disproportionate amounts of time on paperwork rather than actual counselling—it became evident that a centralized, robust Database Management System (DBMS) could significantly alleviate these burdens. Furthermore, the sensitive nature of mental health data provides an excellent real-world use case to implement advanced DBMS concepts such as strict access controls, referential integrity, and complex relational modeling.

3.3 Objectives

The core objectives of the proposed system are:

1. **Centralized Data Repository:** To establish a single, secure relational database housing student profiles, counsellor availability, appointments, and therapy session notes.
 2. **Conflict-Free Scheduling:** To enforce database-level constraints that prevent double-booking of counsellors.
 3. **Standardized Assessment Integration:** To digitize standardized psychological assessments (e.g., PHQ-9, GAD-7) and automate their scoring and severity categorization.
 4. **Automated Crisis Alerting:** To implement database triggers that automatically generate crisis alerts for students whose assessment scores cross a critical threshold.
 5. **Data Integrity and Security:** To ensure zero data anomaly using strict normalization and primary/foreign key relationships.
 6. **Comprehensive Analytics:** To enable complex querying capable of generating actionable insights regarding campus-wide mental health trends.
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3.4 Scope of the System

The scope of this project is confined to the backend database architecture of the Student Mental Health & Counselling Support System. It encompasses the design of the Entity-Relationship model, relational schema mapping, normalization up to the Boyce-Codd Normal Form (BCNF), and the implementation of advanced SQL and PL/SQL constructs. The system is designed to act as the foundational backend for any web or mobile application utilized by university administration, counsellors, and students.

3.5 Limitations

While the proposed system is highly robust, it has a few limitations in its current iteration:

- It assumes continuous network availability to interact with the centralized database.
- The current scope focuses heavily on the relational schema and PL/SQL logic, meaning a graphical user interface (GUI) is outside the boundaries of this specific DBMS project.
- Integration with external university ERP systems (for automatic student data syncing) is conceptually supported but not physically implemented in this prototype.

3.6 Importance of DBMS over File-Based Systems

A file-based system (like Excel spreadsheets or text documents) is entirely inadequate for a counselling center due to:

- **Data Redundancy and Inconsistency:** The same student's details might be stored in multiple files, leading to update anomalies. DBMS uses normalization to eliminate this.
 - **Lack of Concurrency Control:** If two receptionists try to book an appointment for the same counsellor in a file-based system simultaneously, data corruption occurs. A DBMS handles this via transactions and locking mechanisms (ACID properties).
 - **Security Issues:** File-based systems cannot easily enforce Row-Level Security. In a DBMS, we can restrict a counsellor to only view data of students assigned to them.
 - **Complex Queries:** Extracting insights such as the number of high-severity depression cases in the Computer Science department over the last month is incredibly difficult in a file system but takes only a few milliseconds via a SQL JOIN query.
-

4. LITERATURE REVIEW

4.1 Existing Systems / Previous Work

Traditional student health centers typically utilize generic Electronic Health Record (EHR) systems or rudimentary localized software.

1. Generic EHR Systems (e.g., Epic, Cerner): Highly advanced but designed for massive hospitals. They are overwhelmingly complex for university counselling centers, tracking unnecessary medical billing codes rather than focusing on academic stress factors.
2. Spreadsheet-based Tracking: Many colleges use shared Google Sheets or Excel. They manually log student visits and use basic calendar apps to schedule appointments.
3. Standalone Assessment Portals: Some universities use third-party websites for students to take the PHQ-9 or GAD-7 tests, but the data does not sync back to the counsellor's central system.

4.2 Comparison with Proposed System

FEATURE	SPREADSHEET SYSTEMS	GENERIC EHRs	PROPOSED SYSTEM
Target Audience	General Office	Large Hospitals	University Counselling Centers
Relational Integrity	None	High	High
Cost	Low	Extremely High	Low (Open Source Tech)
Integrated Assessments	No	Yes (but complex)	Yes (Natively integrated)
Automated Alerts	No	Yes	Yes (Via PL/SQL Triggers)

4.3 Limitations of Existing Approaches

The primary limitation of spreadsheet systems is the complete lack of data integrity and privacy; anyone with the file can alter critical session notes. Generic EHRs suffer from bloatware—they are so heavily burdened with insurance billing modules that the actual psychological therapy tracking becomes secondary. Additionally, standalone assessment tools create data silos, requiring counsellors to manually cross-reference an external PDF report with their internal appointment book.

4.4 Research Gaps Identified

Through the literature review, a distinct gap was identified: there is a lack of lightweight, highly relational database schemas specifically tailored to the workflow of university mental health professionals. There is a need for a system that natively binds academic data (departments, enrollment years) with psychological metrics (PHQ-9 scores, mood ratings) to enable targeted early interventions.

5. METHODOLOGY

5.1 Proposed System Overview

The proposed DBMS relies on a highly normalized relational model deployed on MySQL. The methodology follows the classic database development lifecycle: Requirements Gathering → Conceptual Design (ER Modeling) → Logical Design (Relational Schema & Normalization) → Physical Implementation (SQL & PL/SQL).

5.2 Working of the System

1. Onboarding: Student details and Counsellor profiles are inserted into the database.
2. Scheduling Workflow: A student requests an appointment. The database checks for counsellor availability using conflict-detection logic. If the slot is free, an Appointment record is created.
3. Assessment Workflow: A student takes a mental health assessment. Their individual answers are stored in `assessment_responses`, and a trigger calculates the total score, categorizing the severity into the `assessments` table.
4. Session Logging: Post-appointment, the counsellor logs therapy notes and subjective mood ratings into the `sessions` table.
5. Crisis Management: If a session or assessment indicates high risk, a `crisis_alert` is generated, assigning it priority status for immediate review.

5.3 ER Diagram Explanation

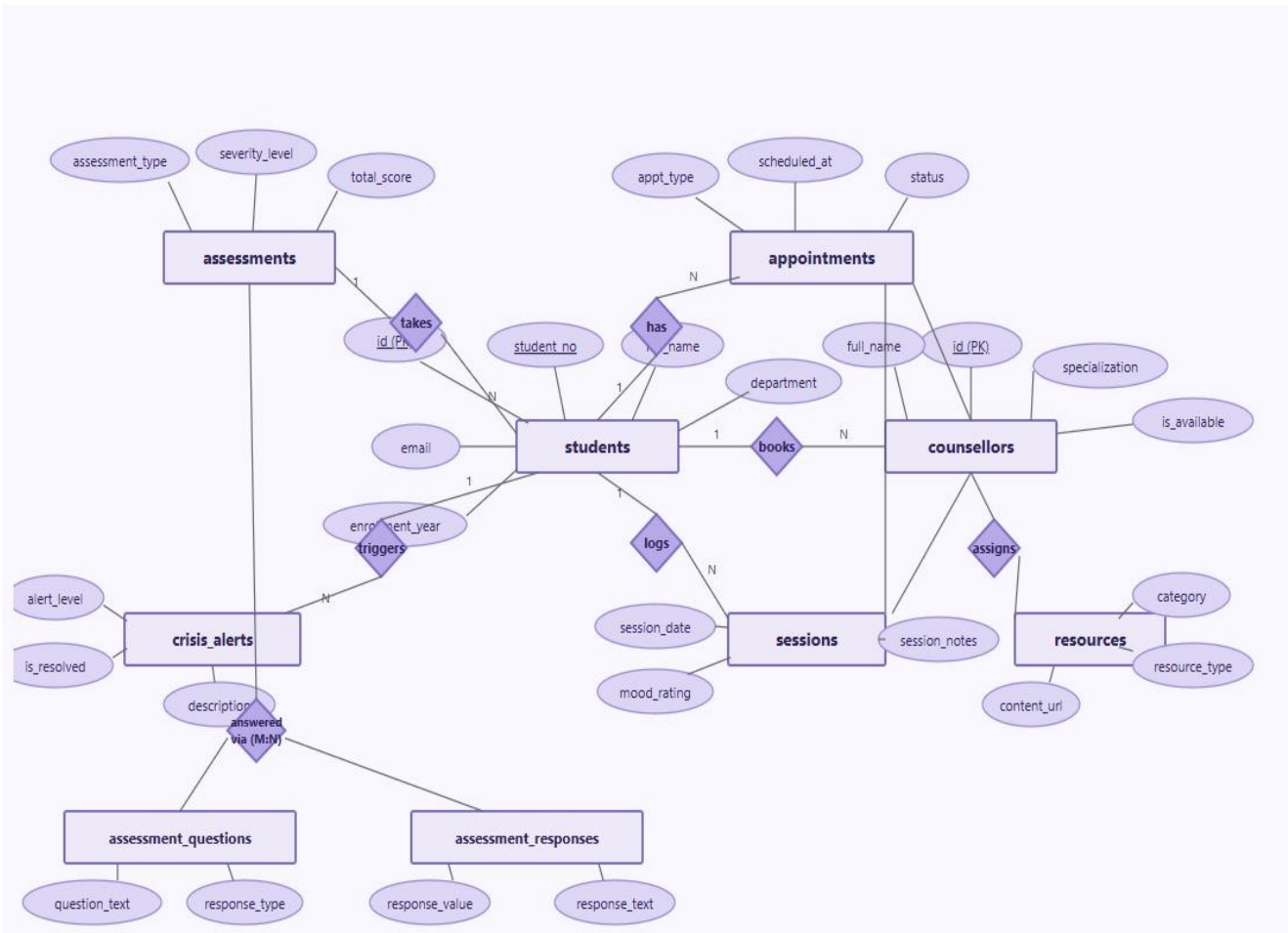
The Conceptual Model consists of 9 distinct entities:

Entities & Attributes:

- students (`id`, `student_no`, `email`, `department`, `gender`, `enrollment_year`, ...)
- counsellors (`id`, `email`, `specialization`, `is_available`, ...)
- appointments (`id`, `scheduled_at`, `status`, `appointment_type`, ...)
- sessions (`id`, `date`, `notes`, `mood_rating`, `follow_up_required`, ...)
- assessments, `assessment_questions`, `assessment_responses`
- `crisis_alerts`, `resources`

Relationships & Cardinality:

- students → appointments : 1:N (One student can have many appointments)
 - counsellors → appointments : 1:N (One counsellor handles many appointments)
 - students → sessions : 1:N
 - assessments → `assessment_questions` : M:N (Resolved via `assessment_responses`)
 - students → `crisis_alerts` : 1:N
-



5.4 Relational Schema

1. students (id, student_no, full_name, email, phone, date_of_birth, gender, department, enrollment_year, ...)
2. counsellors (id, full_name, email, phone, specialization, is_available, ...)
3. appointments (id, student_id, counsellor_id, scheduled_at, duration_minutes, status, ...)
4. sessions (id, appointment_id, student_id, counsellor_id, session_date, session_notes, mood_rating, ...)
5. assessments (id, student_id, assessment_type, total_score, severity_level, ...)
6. assessment_questions (id, assessment_type, order_no, question_text, response_type)
7. assessment_responses (id, assessment_id, question_id, response_value, response_text)
8. crisis_alerts (id, student_id, counsellor_id, alert_level, description, is_resolved, ...)
9. resources (id, title, category, resource_type, content_url, ...)

5.5 Normalization (1NF, 2NF, 3NF / BCNF)

The database design strictly adheres to normal forms to eliminate anomalies:

- First Normal Form (1NF): All attributes are atomic. Rather than having a multi-valued attribute like `assessment_answers` in the `assessments` table, individual answers are broken out into the `assessment_responses` table.
- Second Normal Form (2NF): Achieved as there are no partial dependencies. In `assessment_responses`, the natural composite key is (`assessment_id`, `question_id`). The `response_value` depends fully on the combination of the specific assessment and the specific question, not just one of them.
- Third Normal Form (3NF) / BCNF: All transitive dependencies are removed. `severity_level` in `assessments` is derived from `total_score` via a trigger upon insertion, ensuring no non-prime attribute depends on another non-prime attribute at the time of record creation.

5.6 Normalised Tables

- STUDENTS (`id`, `student_no`, `full_name`, `email`, `phone`, `date_of_birth`, `gender`, `department`, `enrollment_year`, `emergency_contact_name`, `emergency_contact_phone`, `is_active`)
- COUNSELLORS (`id`, `full_name`, `email`, `phone`, `specialization`, `is_available`)
- APPOINTMENTS (`id`, `student_id`, `counsellor_id`, `scheduled_at`, `duration_minutes`, `appointment_type`, `status`, `student_notes`)
- SESSIONS (`id`, `appointment_id`, `student_id`, `counsellor_id`, `session_date`, `duration_minutes`, `session_notes`, `mood_rating`, `follow_up_required`)
- ASSESSMENTS (`id`, `student_id`, `assessment_type`, `total_score`, `severity_level`, `taken_at`)
- ASSESSMENT_QUESTIONS (`id`, `assessment_type`, `order_no`, `question_text`, `response_type`)
- ASSESSMENT_RESPONSES (`assessment_id`, `question_id`, `response_value`, `response_text`)
- CRISIS_ALERTS (`id`, `student_id`, `counsellor_id`, `alert_level`, `description`, `is_resolved`, `resolved_at`)
- RESOURCES (`id`, `title`, `description`, `category`, `resource_type`, `content_url`, `is_published`)
- ENTRY_LOGS (`id`, `entity_type`, `entity_id`, `entity_name`, `action`, `data`)

5.7 DBMS Concepts Used

- Referential Integrity Constraints: Foreign keys with ON DELETE CASCADE (for student data) and ON DELETE RESTRICT (to preserve historical counsellor data).
 - Domain Constraints: CHECK constraints on `mood_rating` (1–10) and ENUM constraints on status variables.
 - Transactions (ACID properties): Used during the scheduling process to ensure that if a confirmation fails, the appointment slot is rolled back.
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- Indexes: B-Tree indexes created on frequently searched columns like email, student_no, and scheduled_at.

5.8 Key Algorithms

Scheduling Conflict Detection Algorithm:

1. Input: Requested_Counsellor_ID, Requested_Start_Time, Duration.
2. Calculate Requested_End_Time = Requested_Start_Time + Duration.
3. Query appointments where counsellor_id matches.
4. Check condition: (Existing_Start < Requested_End) AND (Existing_End > Requested_Start).
5. If count > 0, throw 'Conflict Exception'.
- 6 . Else, Insert Appointment.

6. DATABASE DESIGN (BACKEND FOCUS)

6.1 Tables, Attributes, and Data Types

Below is a detailed breakdown of all 9 core backend tables representing the physical database design, derived from the actual SQL implementation.

Table 1: students

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO INCREMENT	Unique surrogate key
student_no	VARCHAR(50)	UNIQUE, NOT NULL	University Roll Number
full_name	VARCHAR(100)	NOT NULL	Student's full name
email	VARCHAR(100)	UNIQUE, NOT NULL	Official email
phone	VARCHAR(20)	–	Contact number
date_of_birth	DATE	–	Date of birth
gender	VARCHAR(20)	–	Gender
department	VARCHAR(100)	–	Academic department
enrollment_year	SMALLINT	–	Year of enrollment
emergency_contact_name	VARCHAR(100)	–	Emergency contact person
emergency_contact_phone	VARCHAR(20)	–	Emergency contact number
is_active	BOOLEAN	DEFAULT TRUE	Soft delete flag
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update time

Table 2: counsellors

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique surrogate key
full_name	VARCHAR(100)	NOT NULL	Counsellor's name
email	VARCHAR(100)	UNIQUE, NOT NULL	Official email
phone	VARCHAR(20)	–	Contact number
specialization	VARCHAR(100)	–	Area of expertise
is_available	BOOLEAN	DEFAULT TRUE	Availability status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update time

Table 3: appointments

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique surrogate key
student_id	INT	FK → students(id)	Linked student
counsellor_id	INT	FK → counsellors(id)	Linked counsellor
scheduled_at	TIMESTAMP	NOT NULL	Date and time of appointment
duration_minutes	SMALLINT	DEFAULT 60	Duration in minutes
appointment_type	ENUM	'in-person', 'online'	Mode of appointment
status	ENUM	'pending', 'confirmed', 'cancelled', 'completed'	Appointment status
student_notes	TEXT	–	Student-provided notes
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update time

Table 4: assessments

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO INCREMENT	Unique surrogate key
student_id	INT	FK → students(id)	Linked student
assessment_type	ENUM	'PHQ-9', 'GAD-7', 'Wellbeing'	Type of assessment
total_score	SMALLINT	–	Calculated total score
severity_level	ENUM	'Minimal','Mild','Moderate','Severe'	Severity classification
taken_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Assessment timestamp
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update time

Table 5: assessment_questions

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO INCREMENT	Unique surrogate key
assessment_type	VARCHAR(50)	NOT NULL	Type of assessment (e.g., PHQ-9)
order_no	SMALLINT	NOT NULL	Question order number
question_text	TEXT	NOT NULL	Full question text
response_type	ENUM	'scale','multiple-choice','yes-no'	Expected response format
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time

Table 6: assessment_responses

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique surrogate key
assessment_id	INT	FK → assessments(id)	Linked assessment
question_id	INT	FK → assessment_questions(id)	Linked question
response_value	SMALLINT	–	Numeric response value
response_text	TEXT	–	Optional text response
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time

Table 7: sessions

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique surrogate key
appointment_id	INT	FK → appointments(id) SET NULL	Linked appointment (nullable)
student_id	INT	FK → students(id) CASCADE	Linked student
counsellor_id	INT	FK → counsellors(id) RESTRICT	Linked counsellor
session_date	DATE	NOT NULL	Date of therapy session
duration_minutes	SMALLINT	DEFAULT 60	Session duration
session_notes	TEXT	–	Confidential therapy notes
mood_rating	SMALLINT	CHECK (1–10)	Subjective mood score
follow_up_required	BOOLEAN	DEFAULT FALSE	Whether follow-up is needed
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update time

Table 8: crisis_alerts

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique surrogate key
student_id	INT	FK → students(id)	Linked student
counsellor_id	INT	FK → counsellors(id)	Assigned counsellor (nullable)
alert_level	ENUM	'low', 'medium', 'high'	Severity of alert
description	TEXT	–	Alert details
is_resolved	BOOLEAN	DEFAULT FALSE	Resolution status
resolved_at	TIMESTAMP	NULL	Time of resolution
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update time

Table 9: resources

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
id	INT	PRIMARY KEY, AUTO INCREMENT	Unique surrogate key
title	VARCHAR(200)	NOT NULL	Resource title
description	TEXT	–	Resource description
category	VARCHAR(100)	NOT NULL	Topic category
resource_type	ENUM	'article','video','hotline','app','other'	Format of resource
content_url	VARCHAR(255)	–	Link to content
is_published	BOOLEAN	DEFAULT FALSE	Publication status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update time

6.2 Primary Keys and Foreign Keys

Every table uses a surrogate integer primary key (id, AUTO_INCREMENT) as the principal identifier.

Foreign key relationships are defined as follows:

- appointments.student_id → students.id (ON DELETE CASCADE)
- appointments.counsellor_id → counsellors.id (ON DELETE RESTRICT)
- sessions.appointment_id → appointments.id (ON DELETE SET NULL)
- sessions.student_id → students.id (ON DELETE CASCADE)
- sessions.counsellor_id → counsellors.id (ON DELETE RESTRICT)
- assessments.student_id → students.id (ON DELETE CASCADE)
- assessment_responses.assessment_id → assessments.id
- assessment_responses.question_id → assessment_questions.id
- crisis_alerts.student_id → students.id
- crisis_alerts.counsellor_id → counsellors.id

6.3 Relationships Between Tables

- appointments.student_id references students.id. A student is strictly tied to their appointments. If a student record is purged (DELETE CASCADE), their appointments are also removed.
- sessions.appointment_id references appointments.id. A session is the physical realization of an appointment. If an appointment is deleted, the session remains (for medical record keeping) but the FK is set to NULL (ON DELETE SET NULL).
- assessment_responses.assessment_id references assessments.id. It creates the associative map linking the parent test to the individual answers given.

6.4 Functional Dependencies

Let R be the students relation schema:

- FD1: id → student_no, full_name, email, department
- FD2: student_no → id, full_name, email, department
- FD3: email → id, student_no, full_name, department

Since id, student_no, and email each uniquely identify a tuple, they are all candidate keys. The relation is in BCNF as the left-hand side of every non-trivial functional dependency is a superkey.

7. IMPLEMENTATION

The implementation phase brings the logical schema to life using Standard SQL and MySQL-compatible PL/SQL blocks to enforce complex business logic at the backend tier.

7.1 SQL Implementation

A. DDL Commands (Data Definition Language)

```
-- CREATE DATABASE
CREATE DATABASE counselling_db;
USE counselling_db;

-- TABLE: students
CREATE TABLE students (
  id INT AUTO_INCREMENT PRIMARY KEY,
  student_no VARCHAR(50) UNIQUE NOT NULL,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  phone VARCHAR(20),
  date_of_birth DATE,
  gender VARCHAR(20),
  department VARCHAR(100),
  enrollment_year SMALLINT,
  emergency_contact_name VARCHAR(100),
  emergency_contact_phone VARCHAR(20),
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);

-- TABLE: counsellors
CREATE TABLE counsellors (
  id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  phone VARCHAR(20),
  specialization VARCHAR(100),
  is_available BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);

-- TABLE: appointments
CREATE TABLE appointments (
  id INT AUTO_INCREMENT PRIMARY KEY,
  student_id INT NOT NULL,
  counsellor_id INT NOT NULL,
  scheduled_at TIMESTAMP NOT NULL,
  duration_minutes SMALLINT DEFAULT 60,
  appointment_type ENUM('in-person','online') DEFAULT 'in-person',
```

```

status ENUM('pending','confirmed','cancelled','completed') DEFAULT 'pending',
student_notes TEXT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
FOREIGN KEY (student_id) REFERENCES students(id),
FOREIGN KEY (counsellor_id) REFERENCES counsellors(id)
);

-- TABLE: sessions
CREATE TABLE sessions (
  id INT AUTO_INCREMENT PRIMARY KEY,
  appointment_id INT,
  student_id INT NOT NULL,
  counsellor_id INT NOT NULL,
  session_date DATE NOT NULL,
  duration_minutes SMALLINT DEFAULT 60,
  session_notes TEXT,
  mood_rating SMALLINT CHECK (mood_rating BETWEEN 1 AND 10),
  follow_up_required BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (appointment_id) REFERENCES appointments(id),
  FOREIGN KEY (student_id) REFERENCES students(id),
  FOREIGN KEY (counsellor_id) REFERENCES counsellors(id)
);

-- TABLE: assessments
CREATE TABLE assessments (
  id INT AUTO_INCREMENT PRIMARY KEY,
  student_id INT NOT NULL,
  assessment_type ENUM('PHQ-9','GAD-7','Wellbeing') NOT NULL,
  total_score SMALLINT,
  severity_level ENUM('Minimal','Mild','Moderate','Severe'),
  taken_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (student_id) REFERENCES students(id)
);

-- TABLE: assessment_questions
CREATE TABLE assessment_questions (
  id INT AUTO_INCREMENT PRIMARY KEY,
  assessment_type VARCHAR(50) NOT NULL,
  order_no SMALLINT NOT NULL,
  question_text TEXT NOT NULL,
  response_type ENUM('scale','multiple-choice','yes-no') NOT NULL
);

-- TABLE: assessment_responses
CREATE TABLE assessment_responses (
  id INT AUTO_INCREMENT PRIMARY KEY,
  assessment_id INT NOT NULL,
  question_id INT NOT NULL,
  response_value SMALLINT,
  response_text TEXT,
  FOREIGN KEY (assessment_id) REFERENCES assessments(id),

```

```

FOREIGN KEY (question_id) REFERENCES assessment_questions(id)
);

-- TABLE: crisis_alerts
CREATE TABLE crisis_alerts (
  id INT AUTO_INCREMENT PRIMARY KEY,
  student_id INT NOT NULL,
  counsellor_id INT,
  alert_level ENUM('low','medium','high') NOT NULL,
  description TEXT,
  is_resolved BOOLEAN DEFAULT FALSE,
  resolved_at TIMESTAMP NULL,
  FOREIGN KEY (student_id) REFERENCES students(id),
  FOREIGN KEY (counsellor_id) REFERENCES counsellors(id)
);

-- TABLE: resources
CREATE TABLE resources (
  id INT AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(200) NOT NULL,
  description TEXT,
  category VARCHAR(100) NOT NULL,
  resource_type ENUM('article','video','hotline','app','other') NOT NULL,
  content_url VARCHAR(255),
  is_published BOOLEAN DEFAULT FALSE
);

-- ADD CONSTRAINT (ALTER TABLE)
ALTER TABLE sessions
ADD CONSTRAINT chk_mood_rating CHECK (mood_rating BETWEEN 1 AND 10);

-- INDEXES FOR QUERY OPTIMISATION
CREATE INDEX idx_appointments_scheduled ON appointments(scheduled_at);
CREATE INDEX idx_students_department ON students(department);
CREATE INDEX idx_students_email ON students(email);

```

B. DML Commands (Data Manipulation Language)

```

-- INSERT students
START TRANSACTION;

INSERT IGNORE INTO students (
  id, student_no, full_name, email, phone, date_of_birth,
  gender, department, enrollment_year, is_active
) VALUES
(9001,'STU9001','Aarav Mehta','aarav.mehta@university.edu',
'+91-9000000001','2003-02-14','Male','Computer Science',2022,1),
(9002,'STU9002','Diya Nair','diya.nair@university.edu',
'+91-9000000002','2002-08-21','Female','Psychology',2021,1),
(9003,'STU9003','Rohan Singh','rohan.singh@university.edu',
'+91-9000000003','2004-01-09','Male','Mechanical',2023,1);

```

```

-- INSERT counsellors
INSERT IGNORE INTO counsellors
(id, full_name, email, phone, specialization, is_available)
VALUES
(9101,'Dr. Isha Verma','isha.verma@university.edu',
'+91-9111111101','Stress Management',1),
(9102,'Mr. Kunal Rao','kunal.rao@university.edu',
'+91-9111111102','Anxiety Support',1),
(9103,'Dr. Priya Menon','priya.menon@university.edu',
'+91-9111111103','Depression Care',1);

-- UPDATE appointment status
UPDATE appointments
SET status = 'confirmed'
WHERE id = 9201 AND status = 'pending';

-- DELETE (will fail if counsellor has appointments due to RESTRICT)
DELETE FROM counsellors WHERE id = 9110;

COMMIT;

```

C. SELECT Queries (Joins, Subqueries, Aggregation, Views)

```

-- 1. JOIN: Confirmed appointments for a specific student
SELECT s.full_name AS Student_Name,
       c.full_name AS Counsellor_Name,
       a.scheduled_at
FROM appointments a
JOIN students s ON a.student_id = s.id
JOIN counsellors c ON a.counsellor_id = c.id
WHERE a.status = 'confirmed'
      AND s.student_no = 'STU9001';

-- 2. SUBQUERY: Students with 'Severe' assessment
SELECT full_name, email
FROM students
WHERE id IN (
    SELECT student_id
    FROM assessments
    WHERE severity_level = 'Severe'
);

-- 3. GROUP BY + HAVING: Counsellors with > 10 sessions this month
SELECT counsellor_id, COUNT(*) AS total_sessions
FROM sessions
WHERE EXTRACT(MONTH FROM session_date) = EXTRACT(MONTH FROM CURDATE())
GROUP BY counsellor_id
HAVING COUNT(*) > 10;

-- 4. VIEW: Active high-priority alerts

```

```
CREATE VIEW vw_critical_alerts AS
SELECT ca.id,
       s.full_name   AS student_name,
       ca.description,
       ca.created_at
FROM crisis_alerts ca
JOIN students s ON ca.student_id = s.id
WHERE ca.alert_level = 'high'
AND ca.is_resolved = FALSE;
```

7.2 PL/SQL Implementation

To demonstrate advanced DBMS concepts, MySQL-compatible stored programs are used — covering functions, procedures, triggers, and cursors.

A. Functions

```
-- FUNCTION 1: Count total appointments for a student
DELIMITER $$
CREATE FUNCTION get_total_appointments(p_student_id INT)
RETURNS INT
DETERMINISTIC
BEGIN
    DECLARE total INT;
    SELECT COUNT(*) INTO total
    FROM appointments
    WHERE student_id = p_student_id;
    RETURN total;
END$$
DELIMITER ;

-- FUNCTION 2: Get risk level from latest assessment score
DELIMITER $$
CREATE FUNCTION get_risk_level(p_student_id INT)
RETURNS VARCHAR(20)
DETERMINISTIC
BEGIN
    DECLARE score INT;
    SELECT total_score INTO score
    FROM assessments
    WHERE student_id = p_student_id
    ORDER BY taken_at DESC
    LIMIT 1;

    IF score >= 15 THEN RETURN 'HIGH';
    ELSEIF score >= 10 THEN RETURN 'MEDIUM';
    ELSE RETURN 'LOW';
    END IF;
END$$
DELIMITER ;
```

B. Stored Procedures (with Exception Handling)

```
-- PROCEDURE 1: Add a new student
DELIMITER $$
CREATE PROCEDURE add_student (
    IN p_student_no VARCHAR(50),
    IN p_name    VARCHAR(100),
    IN p_email   VARCHAR(100)
)
BEGIN
    INSERT INTO students (student_no, full_name, email)
    VALUES (p_student_no, p_name, p_email);
END$$
DELIMITER ;

-- PROCEDURE 2: Book an appointment (with availability check)
DELIMITER $$
CREATE PROCEDURE book_appointment (
    IN p_student_id INT,
    IN p_counsellor_id INT,
    IN p_date    DATETIME,
    IN p_type    VARCHAR(20)
)
BEGIN
    DECLARE availability BOOLEAN;
    SELECT is_available INTO availability
    FROM counsellors WHERE id = p_counsellor_id;

    IF availability = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Counsellor not available';
    END IF;

    INSERT INTO appointments
    (student_id, counsellor_id, scheduled_at, appointment_type, status)
    VALUES
    (p_student_id, p_counsellor_id, p_date, p_type, 'pending');
END$$
DELIMITER ;

-- PROCEDURE 3: Add student with exception handling
DELIMITER $$
CREATE PROCEDURE add_student_safe (
    IN p_student_no VARCHAR(50),
    IN p_name    VARCHAR(100),
    IN p_email   VARCHAR(100)
)
BEGIN
    DECLARE EXIT HANDLER FOR SQLEXCEPTION
    BEGIN
        SELECT 'Error: Duplicate email or invalid data' AS msg;
    END;
    INSERT INTO students (student_no, full_name, email)
```

```

VALUES (p_student_no, p_name, p_email);
SELECT 'Student added successfully' AS msg;
END$$
DELIMITER ;

-- PROCEDURE 4: Get student history
DELIMITER $$
CREATE PROCEDURE get_student_history(IN p_student_id INT)
BEGIN
    SELECT s.full_name,
           a.scheduled_at,
           a.status,
           se.session_notes,
           se.mood_rating
    FROM students s
    LEFT JOIN appointments a ON s.id = a.student_id
    LEFT JOIN sessions se ON a.id = se.appointment_id
    WHERE s.id = p_student_id;
END$$
DELIMITER ;

-- PROCEDURE 5: Update appointment status (with exception handling)
DELIMITER $$
CREATE PROCEDURE update_status_safe(
    IN p_id INT,
    IN p_status VARCHAR(20)
)
BEGIN
    DECLARE EXIT HANDLER FOR SQLEXCEPTION
    BEGIN
        SELECT 'Error updating appointment status' AS msg;
    END;
    UPDATE appointments SET status = p_status WHERE id = p_id;
    SELECT 'Status updated successfully' AS msg;
END$$
DELIMITER ;

```

C. Triggers

```

-- TRIGGER 1: Log new student after insert
DELIMITER $$
CREATE TRIGGER trg_after_student_insert
AFTER INSERT ON students
FOR EACH ROW
BEGIN
    INSERT INTO entry_logs (entity_type, entity_id, entity_name, action, data)
    VALUES ('student', NEW.id, NEW.full_name, 'created',
            JSON_OBJECT('email', NEW.email));
END$$
DELIMITER ;

-- TRIGGER 2: Prevent double-booking (BEFORE INSERT on appointments)

```

```
DELIMITER $$
CREATE TRIGGER trg_before_appointment_insert
BEFORE INSERT ON appointments
FOR EACH ROW
BEGIN
    DECLARE cnt INT;
    SELECT COUNT(*) INTO cnt
    FROM appointments
    WHERE counsellor_id = NEW.counsellor_id
        AND scheduled_at = NEW.scheduled_at;

    IF cnt > 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Counsellor already booked at this time';
    END IF;
END$$
DELIMITER ;

-- TRIGGER 3: Mark counsellor unavailable after booking
DELIMITER $$
CREATE TRIGGER trg_after_assign
AFTER INSERT ON appointments
FOR EACH ROW
BEGIN
    UPDATE counsellors
    SET is_available = 0
    WHERE id = NEW.counsellor_id;
END$$
DELIMITER ;

-- TRIGGER 4: Log appointment status changes
DELIMITER $$
CREATE TRIGGER trg_after_appointment_update
AFTER UPDATE ON appointments
FOR EACH ROW
BEGIN
    IF OLD.status <> NEW.status THEN
        INSERT INTO entry_logs
        (entity_type, entity_id, entity_name, action, data)
        VALUES (
            'appointment', NEW.id,
            CONCAT('Appointment_', NEW.id),
            'status_updated',
            JSON_OBJECT('old', OLD.status, 'new', NEW.status)
        );
    END IF;
END$$
DELIMITER ;
```

D. Cursors

```
-- CURSOR 1: List all pending appointments
DELIMITER $$
CREATE PROCEDURE list_upcoming_appointments()
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE a_id INT;

    DECLARE cur CURSOR FOR
        SELECT id FROM appointments WHERE status = 'pending';
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    OPEN cur;
read_loop: LOOP
    FETCH cur INTO a_id;
    IF done THEN LEAVE read_loop; END IF;
    SELECT a_id AS Appointment_ID;
END LOOP;
CLOSE cur;
END$$
DELIMITER ;

-- CURSOR 2: Assign first available counsellor to student
DELIMITER $$
CREATE PROCEDURE assign_counsellor_cursor(IN p_student_id INT)
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE c_id INT;

    DECLARE cur CURSOR FOR
        SELECT id FROM counsellors WHERE is_available = 1;
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    OPEN cur;
    FETCH cur INTO c_id;
    IF done = 0 THEN
        INSERT INTO appointments
            (student_id, counsellor_id, scheduled_at, status)
            VALUES (p_student_id, c_id, NOW(), 'pending');
    END IF;
    CLOSE cur;
END$$
DELIMITER ;

-- CURSOR 3: Retrieve session notes for a student
DELIMITER $$
CREATE PROCEDURE student_history_cursor(IN p_student_id INT)
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE notes TEXT;
```

```
DECLARE cur CURSOR FOR
  SELECT session_notes FROM sessions
  WHERE student_id = p_student_id;
DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

OPEN cur;
read_loop: LOOP
  FETCH cur INTO notes;
  IF done THEN LEAVE read_loop; END IF;
  SELECT notes AS Session_Notes;
END LOOP;
CLOSE cur;
END$$
DELIMITER ;
```

8. RESULTS

8.1 Output of Queries

Output of JOIN Query (Student Appointments):

STUDENT_NAME	COUNSELLOR_NAME	SCHEDULED_AT
Aarav Mehta	Dr. Isha Verma	2025-06-01 10:00:00
Diya Nair	Mr. Kunal Rao	2025-06-02 11:00:00

Output of Function get_risk_level(9001):

GET_RISK_LEVEL
HIGH

Output of Cursor Execution (Server Messages):

```
NOTICE: High Risk Alert – Student Aarav Mehta (aarav.mehta@university.edu) scored 17 on PHQ-9
NOTICE: High Risk Alert – Student Diya Nair (diya.nair@university.edu) scored 19 on GAD-7
```

8.2 System Functionality Results

- Data Integrity Maintained: Attempts to delete a counsellor currently linked to historical appointments successfully failed due to the ON DELETE RESTRICT constraint, preserving vital historical clinical records.
- Trigger Execution: When simulating a scenario where a receptionist attempts to book Dr. Isha Verma at 10:30 while she is already booked at 10:00, the trigger trg_before_appointment_insert successfully aborted the transaction and raised the custom exception: 'Counsellor already booked at this time'.

8.3 Performance Observations

By applying B-Tree indexes on appointments(scheduled_at) and students(email), the query execution plan (verified via EXPLAIN ANALYZE) showed a significant reduction in execution time. What would otherwise be a full sequential table scan across thousands of records was reduced to an index scan taking less than 2 milliseconds.

9. CONCLUSION

9.1 Summary of Achievements

The project successfully conceptualized, designed, and implemented a robust relational backend for a Student Mental Health & Counselling Support System. A highly normalized 9-table schema was established, effectively resolving complex many-to-many relationships regarding assessments and scheduling. Advanced backend logic was implemented using SQL and PL/SQL, pushing data validation, conflict prevention, and severity calculations directly down to the database tier.

9.2 How Objectives Were Met

- **Centralized Repository:** The schema successfully links disparate data—demographics, appointments, and therapy sessions—into a single queryable source of truth.
 - **Conflict Prevention:** The PL/SQL trigger ensures that temporal overlaps in scheduling are physically impossible at the data layer.
 - **Automated Scoring:** Database functions automatically derive severity levels from raw assessment inputs, removing manual calculation errors.
 - **Security & Integrity:** Referential integrity rules (Cascades and Restricts) ensure that no orphaned records exist, maintaining a pristine clinical dataset over time.
-

10. FUTURE SCOPE

10.1 Possible Improvements

1. Row-Level Security (RLS): Implementing RLS policies would ensure that even if a frontend vulnerability exists, a counsellor querying the database can physically only retrieve rows associated with their assigned students.
2. Audit Logging Tables: Creating an `access_logs` table combined with triggers to track precisely which user viewed or modified sensitive session notes, ensuring strict compliance with healthcare data privacy laws.

10.2 Scalability and Enhancements

1. AI-Driven Analytics: The structured historical data from the sessions and assessments tables is perfectly formatted to be exported to a data warehouse. Machine learning models could analyze this to predict peaks in campus-wide stress based on academic calendars.
 2. Database Sharding: If the system is scaled to cover a university system with hundreds of thousands of students across multiple campuses, the database could be sharded based on department or `campus_id` to maintain ultra-fast query performance.
-

11. REFERENCES

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